Undergraduate Honors Thesis - Analysis of Strontium Titanate RHEED images using unsupervised learning algorithms

Rahim Raja

University of Chicago - Yang Lab

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Abstract

Strontium Titanate (STO) is a popular choice of substrate to grow 2D materials such as thin film high T_c superconductors such as Iron Selenide films with Molecular Beam Epitaxy. STO, however exhibits different surface reconstructions, at which the efficacy of material growth varies. These surface reconstructions are observed through the patterns of reflection high energy electron diffraction (RHEED), which require a trained eye to accurately and reliably distinguish throughout the annealing process, as the patterns are dependant on the specific system being used as well as many other factors. During this process, the RHEED pattern varies and changes in subtle ways such that an individual with ample experience of characterizing RHEED images are required. We investigate and demonstrate the potential use of feature extraction algorithms on RHEED images to monitor the surface stoichiometry of STO samples. we investigate Principal Component Analysis (PCA) and Non-negative Matrix Factorization (NMF) to be effective methods to process the RHEED images to interpret the sample surface reconstructions of STO (100) during the annealing process.

CONTENTS

I. MOTIVATION AND BACKGROUND

A. Motivation

Strontium Titanate (STO) is a material which is often used as a substrate for thin-film deposition with Molecular Beam Epitaxy. Molecular Beam Epitaxy is a powerful tool for building high quality thin-film materials, such as topological insulators and low T_c super conductors. STO shares similar lattice parameters (3.905 $\mathring{A}$)[1] to many other perovskite crystal structure materials and is thus a commonly used for superconductor thin film growth, as it enables the fabrication of high quality films which do not exhibit interfacial structural disorder.

Preparing an STO substrate for deposition requires the material to be annealed at high temperature. Whilst doing this, the material undergoes many different surface reconstruction transformations, exhibiting combinations of reconstructions such as (1x1), (2x2), ($\sqrt{13} \times \sqrt{13}$), etc[2]. There also exists a diffraction pattern of an unknown surface reconstruction which happens to be a very effective surface reconstruction for thin-film oxide growth. Usually, it requires the expertise of a researcher familiar with RHEED to interpret the many pieces of information inferable in a given RHEED pattern. Since the pattern is also dependant on many variables and is specific to the instrumental setup and state of the system computer vision is nontrivial to implement to computationally understand and characterise the different RHEED patterns[3]. As a result, Principal Component Analysis (PCA) and Non-negative Matrix Factorisation (NMF) are explored as unsupervised learning algorithms to computationally characterize the surface reconstructions of STO samples. Being able to classify different reconstructions will enable greater control of sample properties, including surface reconstruction, such that a researcher can optimise the preparation of a STO sample for the purpose of growing thin-film oxides.

Two similar studied have been previously conducted by Jinkwan Kwoen and Yasuhiko Arakawa at The University of Tokyo. Firstly, they investigated the use of Convolution Neural Network (CNN) and deep learning to create a classification model to identify RHEED patterns of different GaAs surface reconstructions during continuous rotation [4]. This method required surface reconstruction labels to create a successful CNN model. Secondly, the group was able to classify RHEED images of GaAs surface reconstructions during MBE growth using PCA. Since they had labels for the data, however, they were able to measure the effectiveness of their classification model using a confusion matrix [3].

B. Background

1. *Strontium Titanate*

Strontium Titanate, $SrTiO_3$, is a perovskite insulator which is commonly used as a substrate for the growth of superconductors,quantum anomalous hall insulators (such as manganese doped Bi_2Se_3) and oxide-based thin films. It itself becomes superconducting below 0.35K at high electron densities [5]. At room temperature, STO crystallises in the cubic crystal ABO_3 structure with a lattice parameter of 3.905 $\AA$. In this structure, STO has ideal cubic structure such that Ti is at (0,0,0), O is at (0,0,1/2), (0,1/2,0), (1/2,0,0) and Sr is at (1/2,1/2,1/2) in the elementary cell. STO also has a relatively high dielectric constant of 300, which increases further upon cooling [1]. Additionally, doping STO with niobium can make it electrically conductive, which is valuable for perovskite oxide growth since not many conductive single crystals substrates exist. The samples of Strontium Titanate studied in this study are of the (100) crystal face.

STO is also an interesting material to study in microelectronics due to its high storage capacity chemical stability, and its insulating properties and its optical transparency in the visible portion of the electromagnetic spectrum. In general, the tunability of STO makes it a very promising material for future applications, for example. as a photoconductor [6]. To prepare STO for growth, the material must be prepared by ex situ acid etching, and then annealing. The surface is also treated in a furnace with an oxygen environment for several hours to prepare a suitable, uniform surface. After this step, STO is annealed in a molecular beam epitaxy growth chamber before film deposition. This removes residual contaminants, it produces a specific surface reconstruction, and the surface is reduced by creating oxygen vacancies which are responsible for electron doping across the surface. This step is preparing STO, where a surface reconstruction is formed, is not yet standardised and is to some extent, very arbitrary and subjective - it relies on the intuition of the researcher to see the RHEED pattern and decide which surface reconstruction looks favourable. Since different reconstructions have different surface energies, different reconstructions are more conducive for higher quality growths of some films. In addition, some reconstructions, once can affect the interface structure of the grown film and thus potentially affect the properties of the film itself.

A surface reconstruction is the different structure a surface a material forms which is different to its bulk crystal structure, which occurs as the difference forces subject to the atoms on the surface of a material than the bulk, such that the surface energy is minimised. The change in surface reconstruction from the bulk layer structure is described by Wood's notation [7]:

$$X(hkl)m \times n - R\phi$$

, where hkl are the miller indices of the plane (100 for all of the STO samples in our study), m and n are the factors by which the basic translation vectors are scaled by in the surface superstructure, ϕ is the angle which the unit cell is rotated.

For STO, different pre-chamber preparations can result in different initial reconstructions, however, when heated, it cycles through different reconstructions. Understanding this process and understanding how to achieve certain reconstructions can be valuable in characterizing the evolution of surface reconstructions across a surface throughout the annealing process. One of the reconstructions, which is seen at the last panel of figure 6, is referred to as teh the High Temperature Reconstruction (HTR). This is a mysterious reconstruction which is not understood, however, it often one of the best reconstructions to grow superconductors on, for unknown reasons.

2. Molecular Beam Epitaxy

Molecular Beam Epitaxy (MBE) is a method of growing thin films in ultra high vacuum ($10^{-8} to 10^{-12}$ Torr) using beams of atoms emitted from effusion cells which are heated, which react and condense on the crystalline surface of the substrate [8]. A schematic of MBE is shown in figure 1. MBE is a powerful technique which provides great control of materials properties and purity of the growth of thin film materials. The effusion cells, which contain component atoms or molecules, can be shut off using shutter such that individual layers can be created and controlled. By manipulating the pressure and temperature, one can grow very specific 'recipes' for different materials.

This is a technique which can be used both to fabricate materials for research purposes, but also to produce materials for industrial applications, such as semiconductors[7]. The temperature during this process needs to be high such enough atoms on the surface can travel to their lattice sites, but not too high such that diffusion happens between layers of the the bulk material.

The sample inside the MBE chamber can be heated either by radiative heating, where the sample is heated from an external heating component, or through direct current heating, where the sample is heated by flowing a current through it.

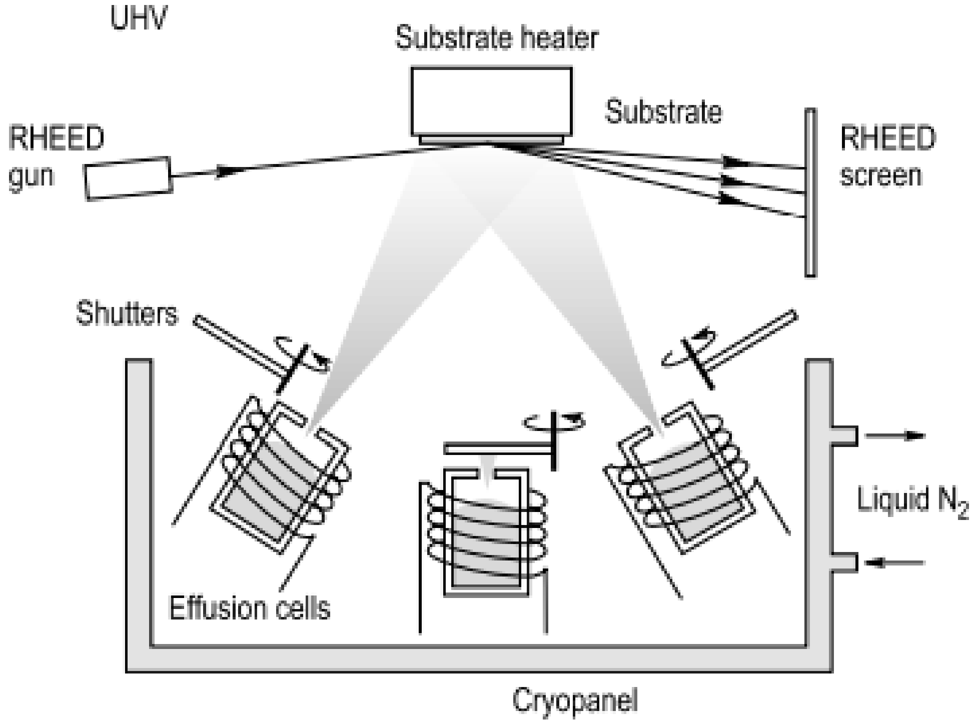

FIG. 1. Sketch of MBE system with RHEED setup. This schematic is taken from 'Surface Science - An Introduction', by Oura K. et Al. [7].

3. *Reflection High Energy Electron Diffraction*

Reflection high-energy electron diffraction (RHEED) is a characterization tool usually implemented into the MBE chamber which uses the diffraction patterns of high energy electrons reflected from the surface (RHEED only probes the surface) of a crystalline sample to infer the physical characteristics of the material, particularly the crystal structure and surface morphology/microstructure [9]. A RHEED setup, as visualised in figure 1, consists of an electron gun which is incident on a clear material surface, and a photo-luminescent detector screen [7]. The electrons strike the surface at a small angle, between 0.5 ° and 2.5 ° [10], diffract with the atoms at the surface of the material, and a proportion of these interfere constructively onto the detector. The electrons can either elastically scatter between the atoms spacing in the surface (kinematic scattering), or they can scatter through multiple diffraction events whilsts losing some energy to the crystal [9] (dynamic scattering). Kinematic scattering is useful for extractive qualitative information about the surface, whilst dynamic scattering is useful extracting quantitative information [10]. Kinematic scattering will be studied here since we are interested in determining qualitative surface details.

The possible reflections are determined by

$$k' - k_0 = G$$

where k' is the diffracted wavevector, k_0 is the incident wave vector and G is the reciprocal lattice vector. For elastic scattering, both incident and diffracted beam vectors must have equal magnitude. To understand the formation of the diffraction patterns, we visualise a sphere in reciprocal space (Ewald sphere) with radius of k_0, and rods of spacing G, which represents the two-dimensional reciprocal lattice (since RHEED only probes the surface of materials). The possible reflections are thus characterised by all $k^{z}\prime$ which connect the origin of the sphere with points of G intersections on the sphere. The This visualised in figure2. Each intersection is projected onto the screen which created 'Laue Circles' of radius L_n centered at H. These Laue circles

The magnitude of the wavevector k_0 is given by

$$k_0 = \frac{1}{h}\sqrt{2m_0E + \frac{E^2}{C^2}}$$

. As a result, for about 10keV electrons (which is what is used in our measurements), the incident wavevector is about $390nm^{-1}$, whilst the reciprocal lattice unit of STO is approximately nm^{-1}. This means that the Ewald sphere is much larger than the spaces between the reciprocal lattice rods and there are multiple intersections. Additionally, since there is a spread of incoming electron energies, the streak we

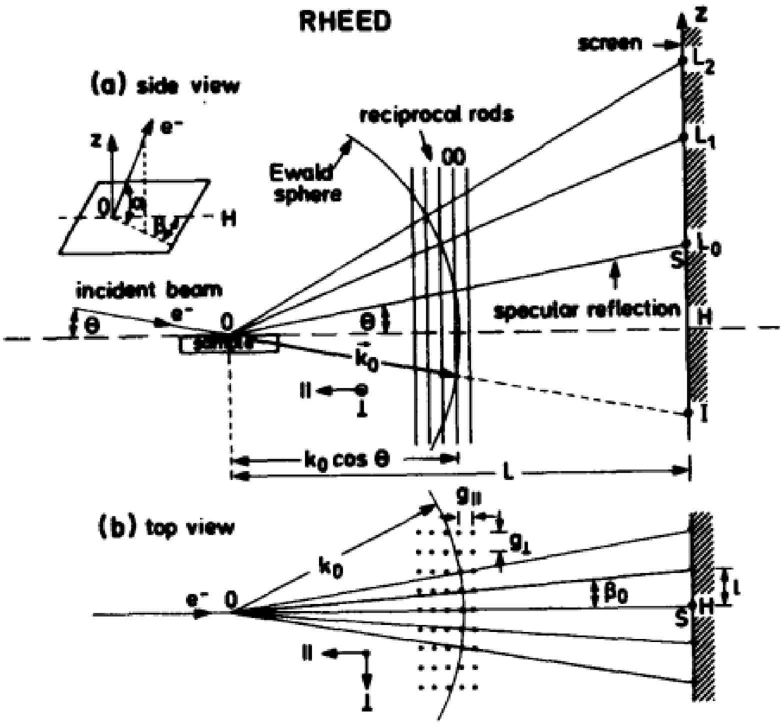

FIG. 2. Diffraction geometry and visualisation of Ewald sphere of RHEED taken from 'Applied RHEED' by Wolfram Braun [10].

observe have a finite thickness since the Ewald sphere becomes a shell with a finite thickness. In addition, the physical structure of the surface impacts the rod shapes and sizes in reciprocal space, which also significantly affects the RHEED pattern. Figure 3, from 'Reflection high-energy electron diffraction study on the SrTiO3 surface structure' by Michio Naito and Hisashi Sato [2] shows the result of different surface morphologies on the streak-like or spot-like behaviour of RHEED diffraction patterns.

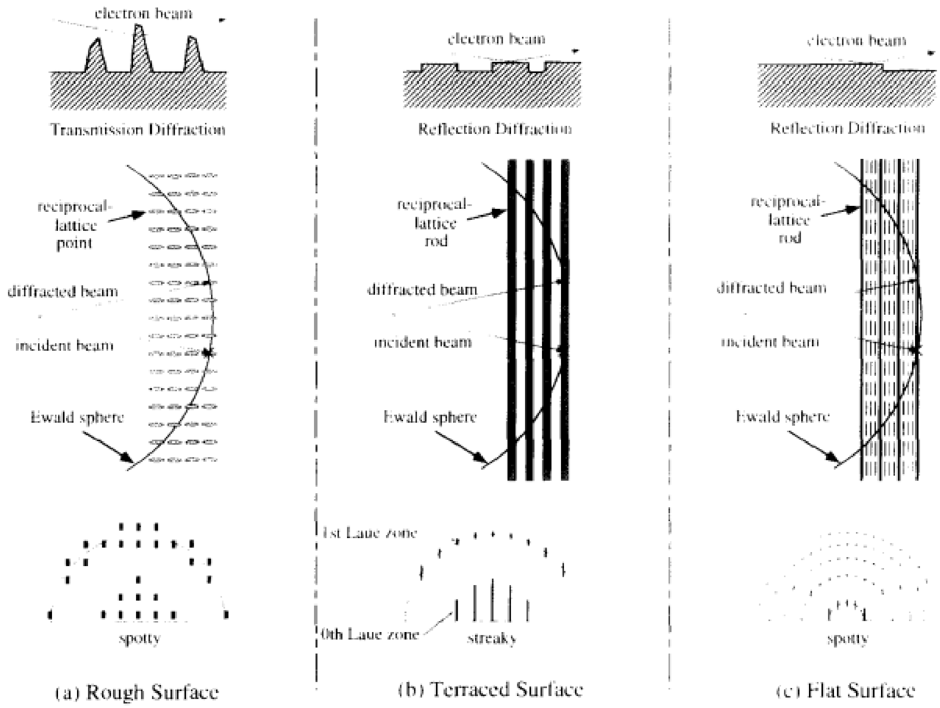

FIG. 3. Illustration showing the relationship between physical surface structure and resulting RHEED patterns. Taken from 'Reflection high-energy electron diffraction study on the SrTiO3 surface structure'. [2]

4. Principal Component Analysis

Principal Component Analysis is a unsupervised learning algorithm. In a real coordinate space with a collection of points, a principal component is a unit vector which points in the direction orthogonal to any other principal components and thus form an orthonormal basis such that the individual dimensions have no linear correlation with each other [11]. Principal components are ordered such that the first principal component is responsible for the most variation in the data, whilst the second principal component is responsible for the most variation in the data when the first is removed, and so on and so forth. Any data point, therefore, can be approximated with a linear combination of the principal components. If more principal components are used, the data gets diminishing closer to the the original data value. PCA is effective in reducing the dimensions of data as it is able to project most of the data point's variation into the first few principal components. As a result, this technique is most effective when the data is highly correlated. With the aid of clustering algorithms, particularly K-Means clustering, PCA is a powerful tool to elucidate such correlations [12].

The principal components themselves can be computed through multiple methods. One example is through singular value decomposition of a data matrix, which is the process of factorizing a square normal matrix with an orthonormal basis to another non-square matrix, as described by

$$M = U\Sigma V^*$$

where M is a $m \times n$ complex matrix, U is a $m \times m$ complex unitary matrix, Σ is an $m \times n$ positive, real diagonal matrix and V is a $n \times n$ complex unitary matrix. By calculating the SVD, one will find the V^* vectors to be the principal components.

Principal components can also be calculated through eigendecomposition of the covariance matrix. The following is a brief outline of the steps involved in this method: first, each feature must be standardised to be a normal distribution with a mean of 0 and variance of 0. This is because PCA favours data with a large variance toward the primary principal components, therefore, without standardisation the data would be skewed to favour the data with inherently larger values. Next, the eigenvectors and eigenvalues are calculated for the covariance matrix.

The covariance matrix is a symmetrical square matrix that gives the covariance (the measure of the direction of relationship between a pair of variables) of the data in two dimensions for each element. For example, in a three-dimensional Cartesian dataset, to get the element in the first column and the third row, one would perform the covariance calculation between the data points' values in the x and z axes. This would be repeated for each element to give the covariance matrix. Next, the eigenvectors and eigenvalues are computed of the covariance matrix. These eigenvectors are the principal components of our data set, with the eigenvectors with the highest eigenvalue being the principal component describing the highest amount of variation in the data set i.e. the first principal component. The number of principal components selected for analysis is subjective and depends on the specific situation. One can plot the cumulative shared variance of the principal components to understand how much variation is captured by what number of principal components.

PCA is a relevant method due to its ease of computation and since it assumes a correlation between features in a data set. Since we will be working with hundreds of images of 656×492 pixels, ease of computation is valuable to iteratively modify and run the model to experiment with different preprocessing techniques and PCA model parameters. To perform PCA on our data, we flatten our images into one dimensional arrays of tuples (of RGB values), and then using each image as a single data point of 322752 dimensions. PCA is performed to return 3 principal components, each again a data point of 322752 dimensions. These data points are then reshaped and constructed into images. These images are 'eigenimages' of the entire dataset, such that their linear combinations can approximately construct each image. We can then plot the coefficients of these eigenimages over the progression of our data collection session to understand the distribution of different principal components and how they change. The limitation of PCA is that the principal components can be difficult to interpret and not necessarily related to distinct physical phenomena. PCA is also sensitive to the scale of features and outliers, so the images' brightness must be scaled appropriately and irrelevant details must be controlled, such as rotation of the sample, so that the diffraction patterns being observed in the principal components relate to physical phenomenon which are relevant to us, namely surface reconstruction.

The second learning technique used is Non-negative Matrix Factorization, which is used to extract common parts of data. This technique is often used for document clustering, missing data imputation, chemometrics, signal processing and of course, computer vision. NMF is a group of multivariate analysis and linear algebra algorithms through which matrices W and H are deduced such that

$$V \approx HW \tag{1}$$

As per the name suggests, each element in the matrix is non-negative. We want to reduce the dimensions of V so that we create a rank r approximation. The value r is equivalent to the number of components we want to reduce our data to. The dimensionalities of the matrices W and H, of $n \times p$ and $p \times m$ respectively, are much smaller than that of the original matrix. Hence, like PCA, NMD is a dimension reduction technique. The method by which NMF is applied to our data is identical to that of PCA. Figure 4 shows the meanings of each matrix equation 1.

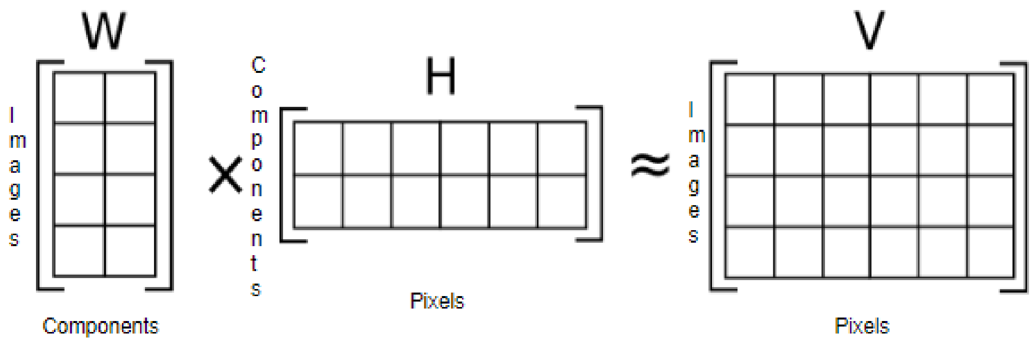

FIG. 4. Visualization of the NMF factorisation. A matrix V is factorised into two matrices with a components dimension, which can be chosen.

The matrix to be factorised, V, is a matrix where each data point is represented through a flattened one dimensional array of tuples of pixels (again, RGB values) through the rows of the matrix. The factors of the matrix consist of H, in which each row represents the components which make up the image (akin to the principal components). This can be split into any amount desired. The W matrix, then, is the proportion of each of these 'eigenimages' present in each data point. For example, image number 7 may consist of $0.3 \times$ eigenimage 1 and $0.5 \times$ eigenimage 2.

NMF was first postulated as a technique viable for parts-based analysis of images by Lee and Seung [13], where they outlined the advantages of using NMF in contrast to PCA for parts-based image analysis. Their primary reasoning for the superiority of PCA is that the constraints of non-negativity of the matrices result in basis images that are additive and not necessarily subtractive, such that they are better representations of the part of images and that the individual components of NMF have more intuitive meaning. Another benefit of NMF is that the image does not to be normalised and corrected as strictly. NMF, similar to PCA, has an inherent clustering property, which makes it a appropriate tool for our purposes [14].

6. Clustering Methods

Finally, since our data is not classified by physical property already (hence why this study must be unsupervised), we may want to explore the possible classifications of our data. After obtaining the relationships between the different components obtained by NMF and PCA, we plot them in 3D to visualise their relationships with each other. The next step is to computationally compute cluster such that we assign labels to images with similar proportions of components. There are two primary methods which were explored for this, K-means clustering[15], and Density Based Spatial Clustering of Applications with Noise (DBScan) clustering [16]. Both of these methods had very similar results, but DBScan was determined to be more appropriate for this dataset.

K-means clustering is a centroid-based algorithm such that all points are separated into K predetermined groups. In order to do this, a K number of centorid are randomly placed. For each data point, the distance to each cluster is calculated, before being assigned a cluster. Finally, the position of the centroids are optimised and the calculations for each data point is reached until the clustering is optimised.

In contrast, DBScan is a density-based algorithm, where the each point must have a minimum number of points in its vicinity. The input parameters are thus ϵ and M, the maximum distance between points to be neighbouring points, and the minumum number of points required to within the ϵ. DBScan is particularly effective when there are outliers and noise in the data. To compute DBScan cluster, each point is classified either as a 'Core Point' or a 'Border Point'. The former is true when there are greater than at least M points in within *epsilon* radius of the point and the latter if there is not M points but another core point within the region. The rest are outlier points, which get eliminated. Core points which are neighbouring are clustered and border points are assigned to each. To determine the ϵ value, we used the 'Knee method', which involves arbitrarily choosing M and computing the average distance of each point to it nearest M neighbours and choose the distance at which there is maximum curvature. M is chosen such that is greater than the number of dimension in our data, and is chosen to be 4 in our analysis.

The advantage of using DBScan is that it doesn't require a predetermined number of clusters. On the other hand, it's parameters do need to be carefully selected and does not work well with clusters with different densities. For our data, DBScan is particularly relevant since the clusters don't necessarily have a tendency to be spherical, which is the case for K-means clustering. Our data is more tendril-like, and thus DBScan is more appropriate. However, it can be useful to impose the number of classes upon our data, for which K-Means clustering is more useful.

II. METHOD AND RESULTS

A. Experimental Procedure

To apply the unsupervised learning methods to STO RHEED data, 3 sets of data was recorded. Each set of data was collected over a period of hours as it was heated to a certain temperature, sustained at a certain temperature for a period of time to anneal it and then slowly cooled. During this annealing process, screenshots of the RHEED diffraction pattern were recorded between 3 or 4 times between each temperature increase. To increase the temperature, we increased the voltage of the heating element on the sample stage. Each increase of voltage of 0.2V corresponded to an increase of 8-12 ° C. In recording this data, the samples were prepared slightly differently such that the initial surface reconstruction, as shown by the RHEED image, was different. In the figures shown, green figures are raw data, and grayscale figures are from data that has been analysed.

Figure 5 shows the RHEED image evolution of the 3 samples as temperature was increased, maintained and then decreased. Sample 1 was prepared by being etched with nitric acid and oxygen annealed at 1000 ° C. It was then heated by resisting heating in ultrahigh vacuuum from 291 ° C to approximately 988 ° C over 8.5 hours and then cooled to 165 ° C over 1 hour. This progression is visualised in figure 5.

In this sample, we see sample start with a combination of 1×1 unreconstructed, $\sqrt{13} \times \sqrt{13}R33.7^\circ$ reconstructed and 2×1 reconstructed. Besides rotation of the sample, we see no change until the sample is cooled to 165 degrees and the $\sqrt{13} \times \sqrt{13}R33.7^\circ$ reconstruction disappears, leaving only the 1×1 unreconstructed and 2×1 reconstructed surfaces.

Sample 2 was 0.05 percent Niobium doped STO, which was heated with direct current heating from 165 ° C to approximately 1064 ° C over 4 hours then cooled to 165 degrees over 2 hours, as represented in figure 6.

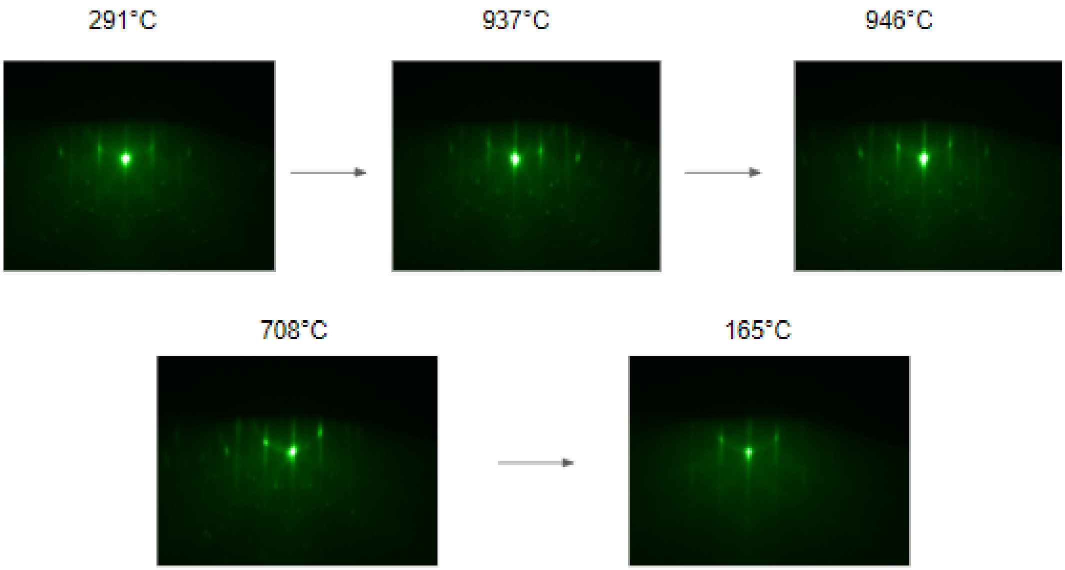

FIG. 5. RHEED image progression of sample 1 throughout the annealing process.

In this sample, we see a faint 1×1 unreconstructed gain in intensity, until the temperature reached 909 degrees Celsius, at which point we see some $\sqrt{13}\times\sqrt{13}R33.7°$ reconstructed. After this point, we notice some other complex reconstruction replace the $\sqrt{13} \times \sqrt{13}R33.7°$. This is characterized by indistinct suborder streaks. This is the mysterious reconstruction which has been experimentally experienced to be ideal for growing superconducting samples.

Sample 3 was also prepared in the same way as the second. Sample 3 was heated from 180 ° C to approximately 950 ° C over 4 hours, kept at 950 ° C for 30 minutes, and then cooled to 165 degrees over 2 hours. This is shown in figure 7.

In this annealing process, we begin with a 1×1 unreconstructed surface, which gains $\sqrt{13}\times\sqrt{13}R33.7°$ and some suborder reconstructed features around 850 degrees Celsius. Upon further heating the $\sqrt{13} \times \sqrt{13}R33.7°$ reconstruction fades at 861 degrees before becoming very apparent in the cooling process at 576 degrees. At this point, the sub order streaks and unreconstructed streaks are also present. These

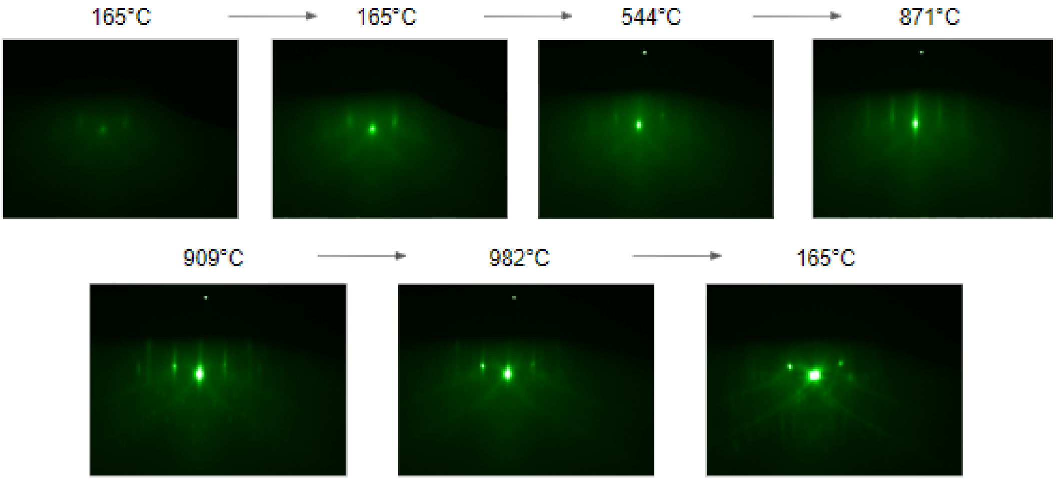

FIG. 6. RHEED image progression of sample 2 throughout the annealing process.

features also intensify by 283 degrees.

There were issues in the first and second data acquisition sessions. Firstly, due to the thermal expansion of the sample, there was a slight rotation of the RHEED angle. As a result, this caused an inconsistent variable in the data in the form of rotation of the RHEED pattern. In the analysis stage, this was very evident and problematic. Since the PCA and NMF algorithms cannot distinguish between if the changes in an image is caused by sample rotation or surface reconstruction change. As a result, the sample rotation was monitored and adjusted closely in the following data collection.

B. Computational Analysis

1. *Data Preprocessing*

Due to fluctuations in brightness on the images, it was necessary to preprocess the data such that the PCA and NMF algorithms would not be affected by anomalous variations in the data. The preprocessing of the data was split into 3 sections:

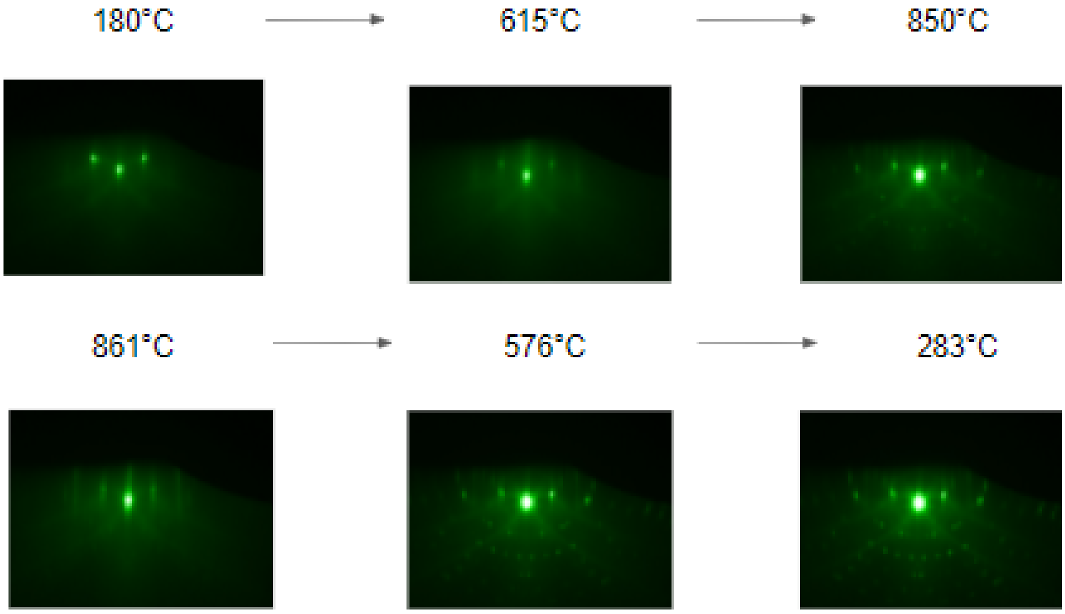

FIG. 7. RHEED image progression of sample 3 throughout the annealing process.

contrast, brightness, and standard scaling.

In addition to these modifications, a black square was added to the center of each RHEED image, as it was found to increase in brightness due to an increase of electron reflectivity. As we are not interested in this phenomenon, it was erased as much as possible without interfering with other features such that the variability in the features due to differing surface reconstructions was accentuated. An image from the 3rd sample data set after it has been processed is shown in figure 8.

The contrast in the images were increased such that there was more definition in the primary diffraction patterns and the background brightness was minimised, since the background brightness is not indicative of any material property which we are interested in. The brightness was normalised by summing the average brightness (pixel value between 1 and 255) of each image in the data set, dividing by the number of images, and finding the difference between this number and each average brightness

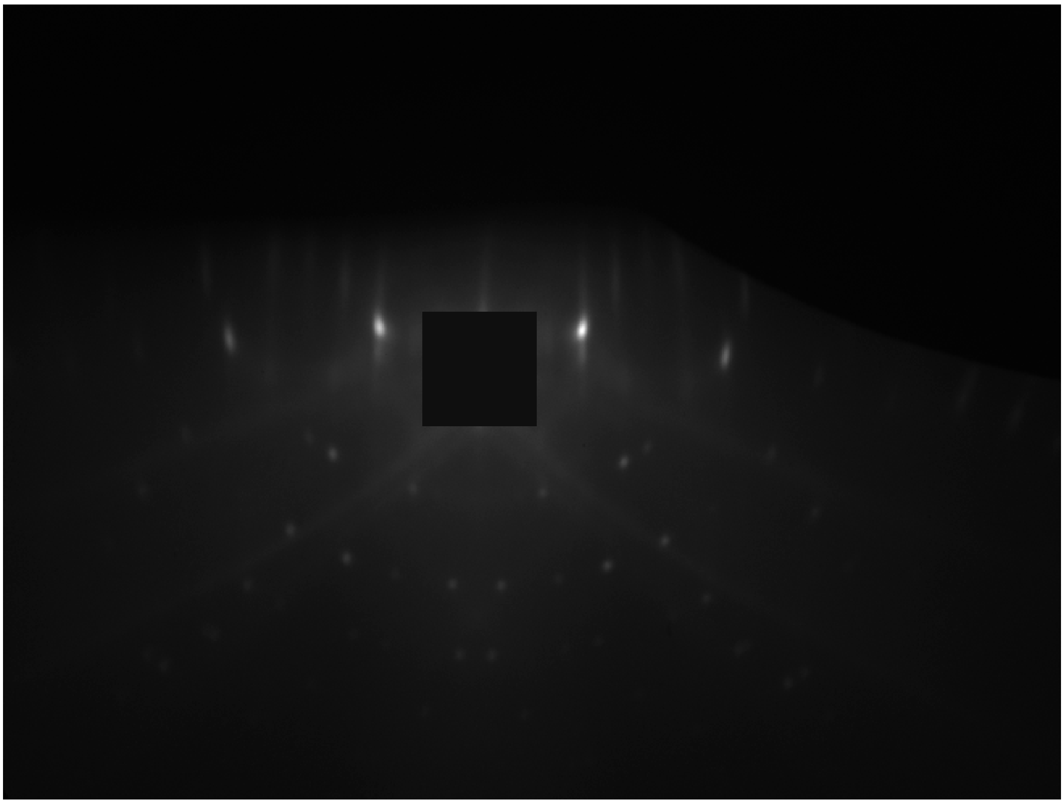

FIG. 8. The 301st Image in the 3rd sample data set - data captured on April 11th 2022.

of each image. This difference, called the 'delta', for each image was then added on to each pixel of each image. Figure 9 depicts an example of how the delta varies across each image through the third sample.

The Python PIL library was used increase contrast in the image and manipulate pixels in the images. Deciding a value for the amount to increase contrast by was arbitrary. It was done such that images regularly spaced out in the data set were increased in contrast by a range of values to visually decide which scale factor resulted in images with clearer diffraction patterns but without a significant feature loss. Figure 10 shows a sequence of images which are increased in brightness and contrast by varying amounts. It was arbitrarily decided that increasing the contrast by a factor of 3 was a satisfactory compromise between feature definition and loss of

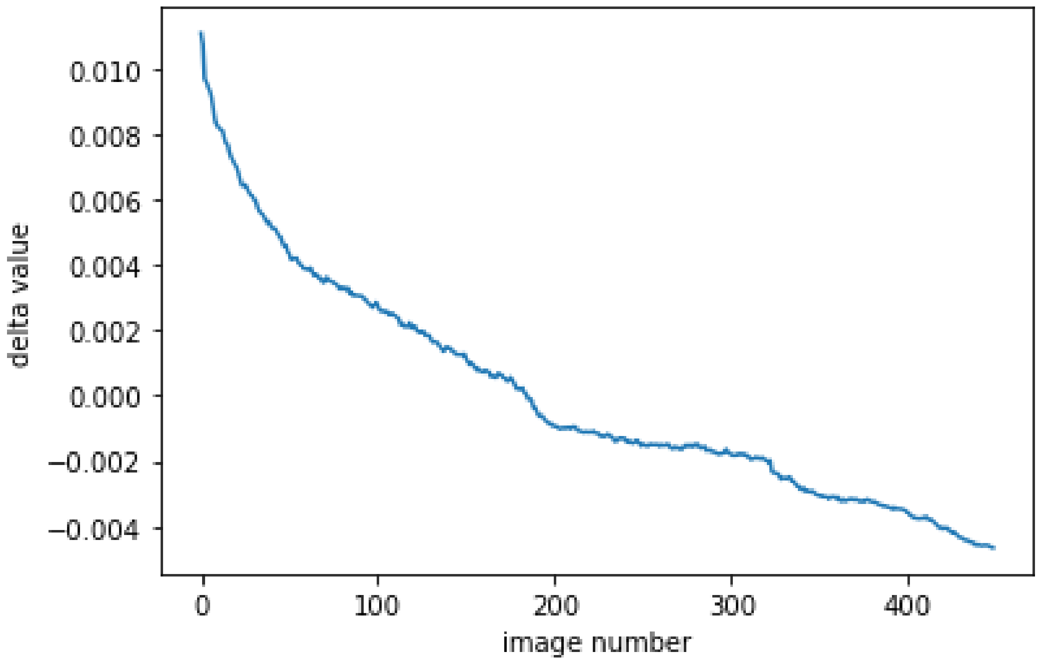

FIG. 9. The variation of brightness from the average brightness of each image. These 'delta' values are then used to adjust the overall brightness of each image such that each image have uniform brightness.

information.

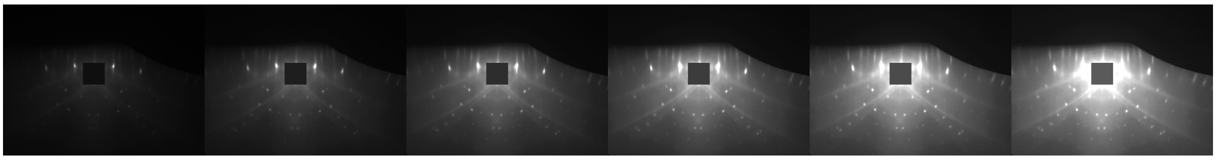

FIG. 10. The 301st Image in the 3rd sample data set with varying increases of contrast (top row) and brightness (bottom row). Here, the leftmost image is the raw image from RHEED.

Standard scaling is a necessary preprocessing stop for many machine learning algorithms, including PCA. This entails rescaling each numerical value such that the data has a mean of 0 and a standard deviation of 1. This is important since PCA will highlight features with high variance and lead to false results. Therefore to eliminate this, we transform our data into a standard normal distribution. This is done using 'Scikit learn'.

'Scikit learn' was generally used to apply PCA and NMF algorithms to our data. Since these techniques require two dimensional input data, our 2 dimensional images were flattened into a one dimensional array of red, blue, green pixel value arrayys. The first dimension represents an individual data point (for example, 449 images), whereas the second represents a single long array of pixel values. PCA and NMF were performed on the two dimensional matrices by creating a model with specified parameters (such as the number of components), fitting this model to a 'pandas' (python library) data frame of the data and then applying the dimensionality reduction on the data to return the transformed data. The number of principal components were chosen based on how much variation of the of the data was cumulatively explained by the principal components. For example, for the sample 3, almost 90 percent of the data set was explained by using 3 components, as shown in figure 11.

The returned data for PCA is an array of the specified number of components with a weight for each image. Each component is a flattened 'eigenimage', which can be reshaped into a 2 dimensional matrix and visualised. For NMF, the output is a pair of matrices, 'W' and 'H', which are approximate factors of the original matrix 'V', as visualised in figure 4.

To interpret the results, the components were plotted against the image number to show the evolution of the relative proportions of components present in the images. In addition, when the algorithms were applied with 3 components, the components were plotted against each other on a 3D plot. By doing this, we can visualise the dependency of different components with each each other. This type of plot shows a clear trajectory of the components of images as they evolve through the annealing process.

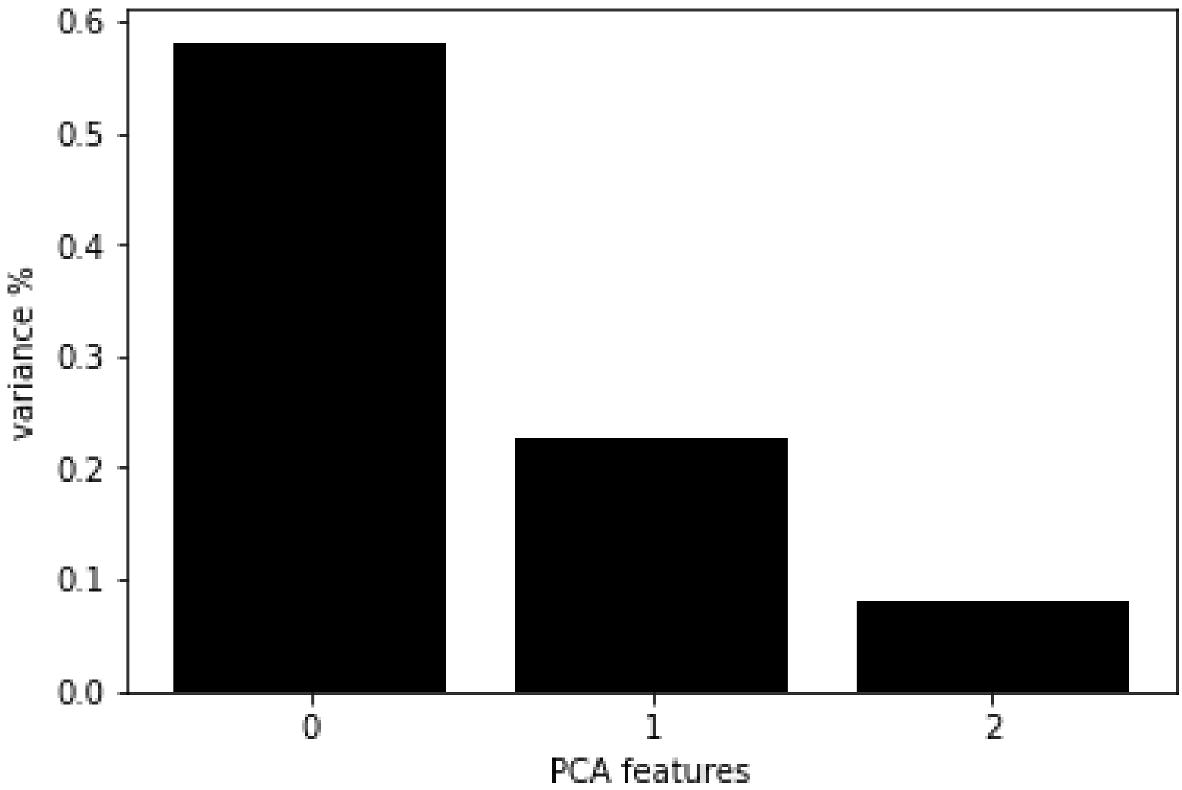

FIG. 11. The percent variance explained by the 0th, 1st and 2nd principal components

2. Application of PCA

In this analysis, the results of the third sample's analysis will be presenting as the first and second sample exhibited anomalous variations in sample rotations and image illumination which made analysis ;ess clear in showing the concepts of PCA and NMF application. We will begin by looking at the third sample's PCA analysis first, as it is the data set with the least irrelevant variation. For the third data set, as shown in figure 11, 90 percent of the data is covered by 3 principal components. The images formed by these principal components are shown in figure 12.

To visualise the principal components, there are 2 different informative ways.

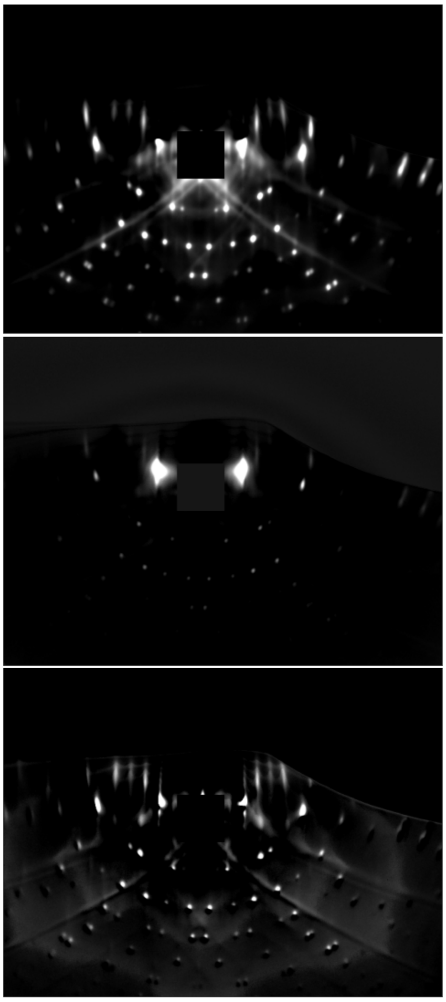

FIG. 12. The 3 principal components as images for sample 3. The top, middle, and bottom image explain 58 percent, 23 percent, and 8 percent of the variation in the images.

Firstly, We may plot the coefficient of each principal component present for each image against other principal components to visualise the relationships between principal components. The latter method is favourable and traditional for implementing clustering methods. In figure 13, we see the evolution of the components across the images taken for the third sample.

We can see in figure 13 that the 0th component generally increases until around

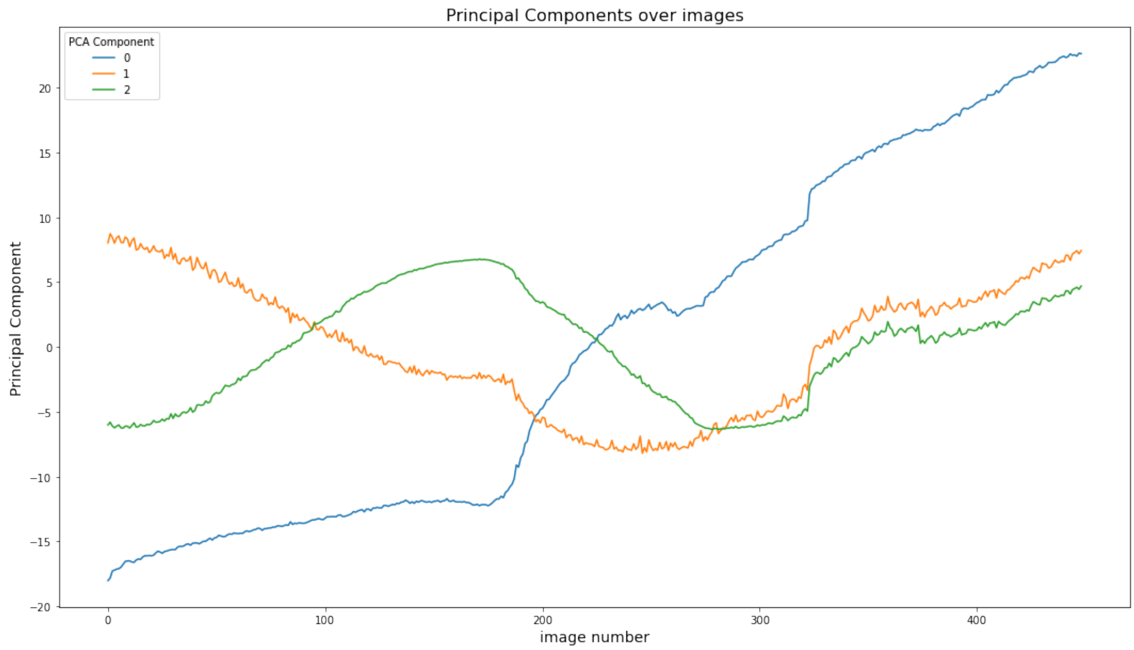

FIG. 13. Evolution of the coefficients of principal components throughout the annealing process.

image just under image 200, where the rate of increase of this component is larger. This makes sense when considering the $\sqrt{13} \times \sqrt{13}R33.7°$ character of the 0th component. The second component, which shows the $1{\times}1$ reconstruction is large near the beginning of teh data and falls of towards the end. This method is evidently an effective way to characterise the proportions and evolution of reconstructions during the annealing process.

We show the results of the relationship between components in figure 14. The top figure, the relationship between principal components coefficients show that through the succession of images, the there is initially little variation of coefficient 2 and then later, little variation in the coefficient 1.

We can then apply both the K-means and DBScan clustering methods to label the data into two sets. The two sets are partitioned around the 322nd or 323rd

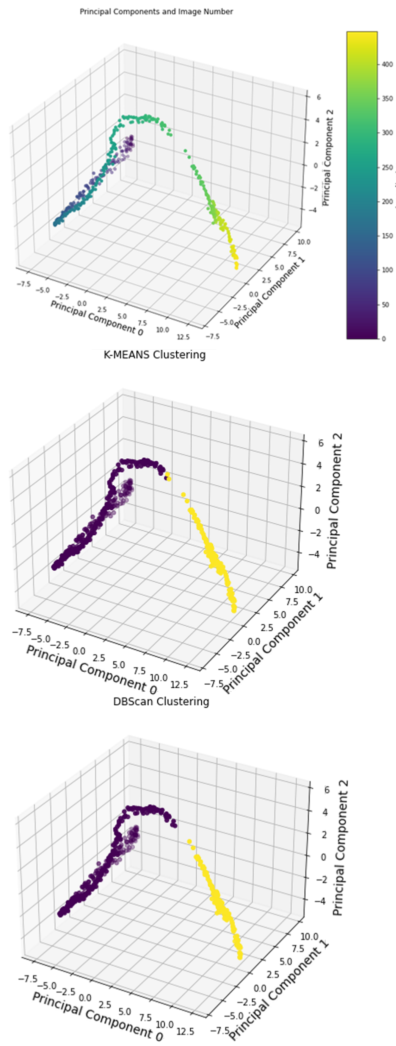

FIG. 14. Top: Principal component coefficients with relation to image number. Center: Principal component coefficients with K-Means clustering. Bottom: Principal component coefficients with DBScan clustering

image. Since the change between each image is so subtle, as multiple images were taken a minute there is no visually perceptible difference between these images at this point, the main difference in data before and after this point is an intensification of the $\sqrt{13} \times \sqrt{13}R33.7°$ surface reconstruction diffraction pattern and suborder peaks.

This comparison is shown in figure 15.

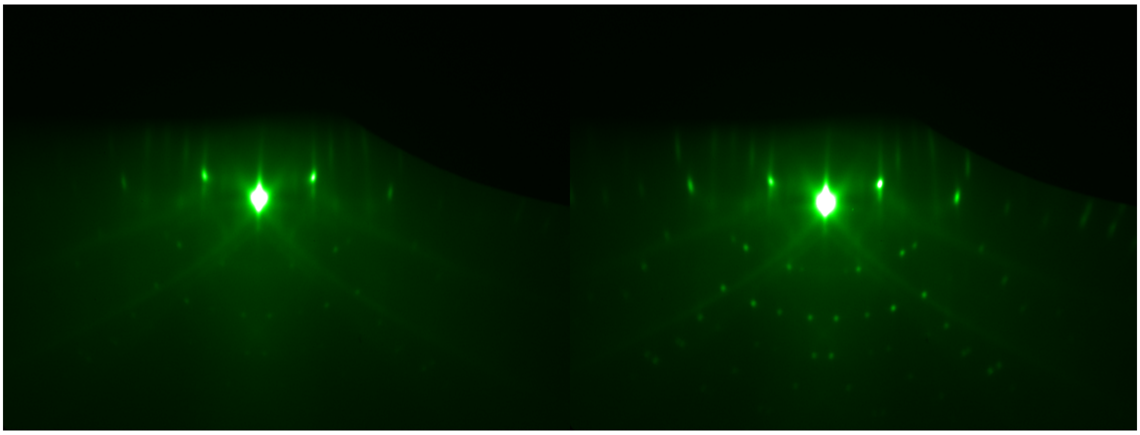

FIG. 15. Comparison of the 283rd image and the 373rd image in the data set for sample 3. The later image has more intense diffraction patterns.

This clustering does not discriminate between reconstructions. This is a weakness of PCA as a method to analyse RHEED diffraction images, as the variation of intensity was favoured over the change of reconstruction. Regardless, the physics shown here is an increase of the surface reconstructions. Finally, for this sample, PCA was performed without preprocessing. This resulted in an output that was much different. Firstly, the principal component images looked much different and had lot more weight towards the first principal component, 76 percent, as opposed to 59 percent. As shown in figure 16, the first and second image have a lot of $\sqrt{13} \times \sqrt{13}R33.7°$ character, with kikuchi lines and some sub order streaks.

The third image mostly depicts background brightness with strong $1{\times}1$ character. Whilst these principal components are not ideal representations for what we are looking for, this analysis is useful for the effect of K-means clustering on this data. As well, as the previous classification determined with the preprocessed data, K-means clustering is able to distinguish a a difference in clusters around image 202. Figure 17 shows the 3-dimensional principal component plot with clusters.

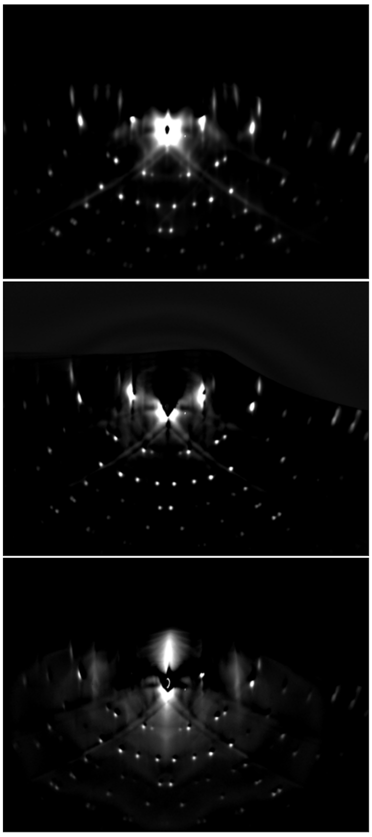

FIG. 16. The 3 principal components as images for sample 3 (without preprocessing). The top, middle, and bottom image explain 76 percent, 11 percent, and 7 percent of the variation in the images.

The difference between the images around image 202, such as image 195 and image 205, are very apparent. In figure 18, one can easily see that this point is the transition between 1×1 reconstruction and the beginning of some $\sqrt{13}\times\sqrt{13}R33.7^{\circ}$ character.

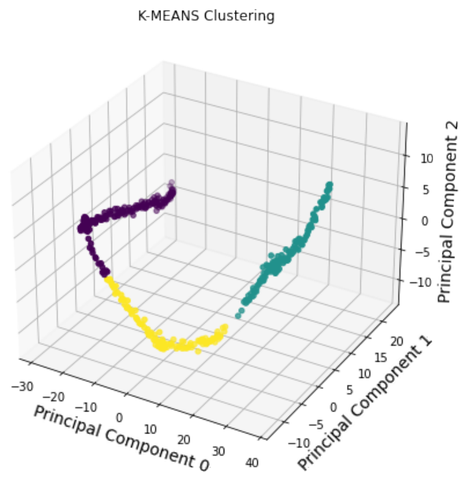

FIG. 17. Principal component coefficients with K-Means clustering.

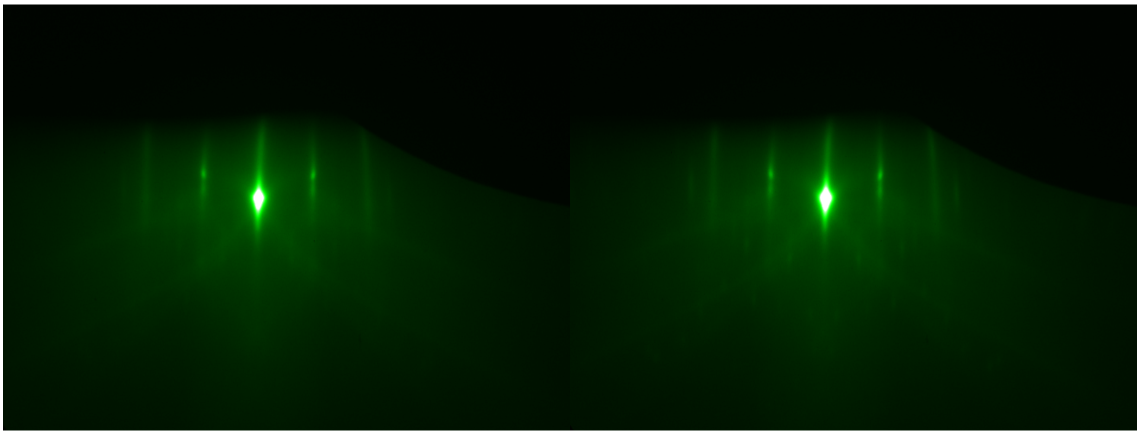

FIG. 18. Comparison of the 195th image and the 205th image in the data set for sample 3. The later image begins to exhibit $\sqrt{13} \times \sqrt{13}R33.7°$ spots.

3. Application of NMF

Next, we shall examine the application of NMF on our data. By applying NMF, we get the plot in figure 19. As opposed to PCA, we get very clear separations of different reconstructions, where the first image is a clear unreconstructed surface RHEED pattern, the second image has $\sqrt{13} \times \sqrt{13}R33.7°$ surface reconstruction, whilst the last also has $\sqrt{13} \times \sqrt{13}R33.7^{\circ}$ reconstruction in addition to stronger suborder streaks.

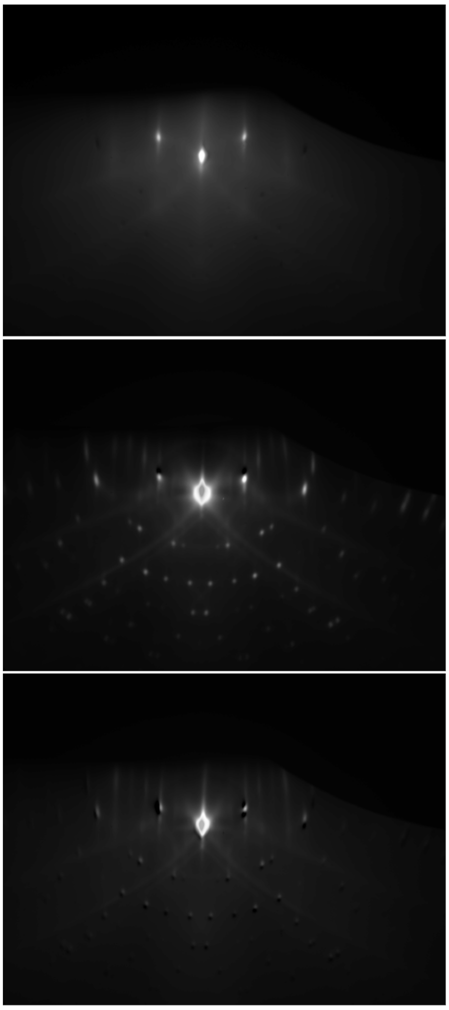

FIG. 19. The 3 NMF components as images for sample 3. Brightness is increased for presentation purposes.

The coefficients of these proportions across the annealing process is presented in figure 20. It is interesting to note that by comparing this figure with figure 13, the 0th component of PCA analysis shares the same form as 1st component of NMF, and the 1st component of PCA shares the same form as the 0th component of NMF.

Once again, the components depict notable changes in trajectory at approximately image number 190 and image number 322. These points are the points where there are indeed notable changes in the surface reconstruction of our samples. These key points of surface reconstruction change are also reflected by the clustering algorithm, where the clusters are separated into different clusters at precisely the same regions in the annealing process for both K-means clustering and DBScan clustering. The plots of the components against each other with clustering is shown in figure 20.

III. DISCUSSION

We have shown the applicability of unsupervised learning methods to characterise the RHEED diffraction patterns of STO during the annealing process. These methods are powerful for their capability of being paired with clustering methods to predict STO surface reconstructions and understand the materials science of how STO (and other materials) change. In addition, by gathering more data from additional data acquisition sessions, the relationship between the PCA or NMF components with temperature may help elucidate how to effectively elucidate how useful reconstructions, such as $\sqrt{13} \times \sqrt{13}R33.7^{\circ}$, are recreated. Furthermore, incorporating these techniques by feeding images into the algorithm and plotting components may be promising in providing more feedback control of sample preparation. A significant difficulty, however, with this method is its sensitivity to drifting in the images, which the algorithm struggles to find effective results for. This issue could be resolved with rotational and brightness corrective image processing.

Whilst we explored the use of unsupervised learning algorithms due to the lack of labelling in our data. Since there is a directionality of whether temperature was being increased and decreased during the annealing process, there is no one-to-one correspondence between temperature label and RHEED pattern. As a result, it would be difficult to effectively apply machine learning algorithms to our data, particu-

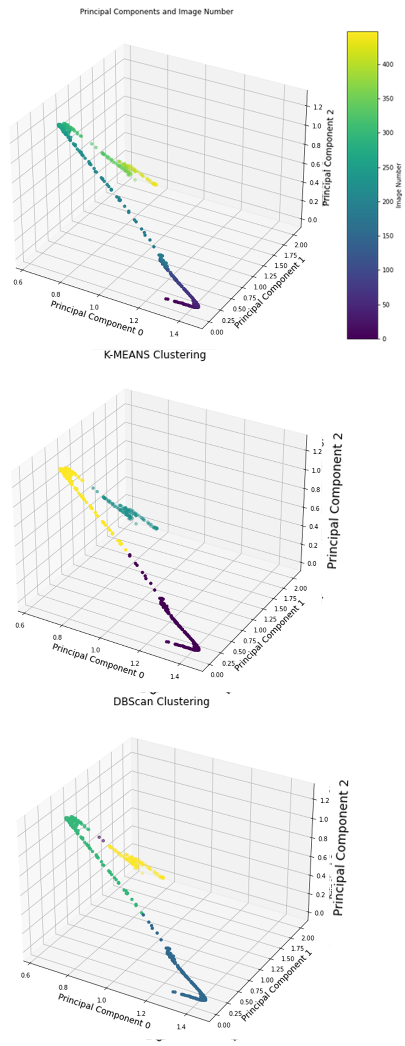

FIG. 20. Top: NMF component coefficients with relation to image number. Center: NMF component coefficients with K-Means clustering. Bottom: NMF component coefficients with DBScan clustering

larly due to the lack of samples. An issue with this method is that the diffraction pattern is dependant on being characterized and labelled by an expert. Even then, some combinations of diffraction patterns cannot be sufficiently described by eye. Therefore, using simulated RHEED data to create diffraction patterns with known percentages of surface reconstructions could enable supervised machine learning techniques to work. Whilst RHEED simulations exist[17], it is difficult to precisely and realistically recreate all the aspects of a true RHEED image with the consideration of inelastic scattering. With an accurate simulation, one could feed a large amount of labelled RHEED images to a convolutional neural network to build a effective model for RHEED analysis.

since the biggest hindrance to performing supervised machine learning was the lack of data, collecting RHEED images of STO (100) at different temperature will enable the use of neural networks to recognise RHEED diffraction features more with greater accuracy. It was shown in this study that PCA and NMF algorithms are effective in understanding the evolution of different aspects of RHEED images during the annealing process. In addition, as described by Dosovitskiy et Al [18], it may already be possible to use neural networks to investigate RHEED images with by augmenting data to create 'surrogate classes', which are classes created by introducing modifications, such as translation, rotation, scaling, contrast, etc. to existing images.

Looking forward, the input parameters of these techniques could be further optimised and modified to extract constituents of data which may correspond more closely to the surface reconstructions, rather than the combination of different reconstruction that our application of the unsupervised learning techniques produced. This was achieved to some extent by the use of NMF, however, it may be possible to modify the algorithms such that they respond more precisely to specific surface reconstructions.

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